



**SCHOOL OF
COMPUTING**

Design Analysis of Algorithm

CSE211

Week-7

CH.SC.U4CSE24221

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1 . Quick sort by different pivots and quick select

CODE:

```
C quicksort.c > main()
1 // CH.SC.U4CSE24221
2 // Implement Quick Sort using three different pivot choices: first element, middle element, and last element
3
4 #include <stdio.h>
5
6 void swap(int *a,int *b)
7 {
8     int t=*a;
9     *a=*b;
10    *b=t;
11 }
12
13 int partitionLast(int arr[],int low,int high)
14 {
15     int pivot=arr[high];
16     int i=low-1;
17
18     for(int j=low;j<high;j++)
19     {
20         if(arr[j]<pivot)
21         {
22             i++;
23             swap(&arr[i],&arr[j]);
24         }
25     }
26
27     swap(&arr[i+1],&arr[high]);
28     return i+1;
29 }
30
31 int partitionFirst(int arr[],int low,int high)
32 {
33     int pivot=arr[low];
34     int i=low+1;
35     int j=high;
36
37     while(1)
38     {
39         while(i<=high && arr[i]<=pivot) i++;
40         while(arr[j]>pivot) j--;
41
42         if(i>=j) break;
43
44         swap(&arr[i],&arr[j]);
45     }
}
```

```
46
47     swap(&arr[low],&arr[j]);
48     return j;
49 }
50
51 int partitionMiddle(int arr[],int low,int high)
52 {
53     int mid=(low+high)/2;
54     swap(&arr[mid],&arr[high]);
55     return partitionLast(arr,low,high);
56 }
57
58 void quickSortLast(int arr[],int low,int high)
59 {
60     if(low<high)
61     {
62         int pi=partitionLast(arr,low,high);
63         quickSortLast(arr,low,pi-1);
64         quickSortLast(arr,pi+1,high);
65     }
66 }
67
68 void quickSortFirst(int arr[],int low,int high)
69 {
70     if(low<high)
71     {
72         int pi=partitionFirst(arr,low,high);
73         quickSortFirst(arr,low,pi-1);
74         quickSortFirst(arr,pi+1,high);
75     }
76 }
77
78 void quickSortMiddle(int arr[],int low,int high)
79 {
80     if(low<high)
81     {
82         int pi=partitionMiddle(arr,low,high);
83         quickSortMiddle(arr,low,pi-1);
84         quickSortMiddle(arr,pi+1,high);
85     }
86 }
87
88 void printArray(int arr[],int n)
89 {
```

```

87
88  void printArray(int arr[],int n)
89  {
90      for(int i=0;i<n;i++)
91          printf("%d ",arr[i]);
92      printf("\n");
93  }
94
95  int main()
96  {
97      printf("CH.SC.U4CSE24221\n");
98
99      int n;
100     scanf("%d",&n);
101
102     int arr1[n],arr2[n],arr3[n];
103
104     for(int i=0;i<n;i++)
105     {
106         scanf("%d",&arr1[i]);
107         arr2[i]=arr1[i];
108         arr3[i]=arr1[i];
109     }
110
111     quickSortFirst(arr1,0,n-1);
112     quickSortMiddle(arr2,0,n-1);
113     quickSortLast(arr3,0,n-1);
114
115     printArray(arr1,n);
116     printArray(arr2,n);
117     printArray(arr3,n);
118
119     return 0;
120 }

```

OUTPUT

```

kantatejesh@KANTAs-MacBook-Air TASK7 % gcc quicksort.c -o quicksort
kantatejesh@KANTAs-MacBook-Air TASK7 % ./quicksort
CH.SC.U4CSE24221
5
8 3 5 6 4
3 4 5 6 8
3 4 5 6 8
3 4 5 6 8
kantatejesh@KANTAs-MacBook-Air TASK7 % 

```

2. QUICK SELECT

CODE:

```
C quickselect.c > swap(int *, int *)
1 // CH.SC.U4CSE24221
2 //Quick Select algorithm
3
4 #include <stdio.h>
5
6 void swap(int *a,int *b)
7 {
8     int t=*a;
9     *a=*b;
10    *b=t;
11 }
12
13 int partition(int arr[],int low,int high)
14 {
15     int pivot=arr[high];
16     int i=low;
17
18     for(int j=low;j<high;j++)
19     {
20         if(arr[j]<=pivot)
21         {
22             swap(&arr[i],&arr[j]);
23             i++;
24         }
25     }
26
27     swap(&arr[i],&arr[high]);
28     return i;
29 }
30
31 int quickSelect(int arr[],int low,int high,int k)
32 {
33     if(low<=high)
34     {
35         int pi=partition(arr,low,high);
36
37         if(pi==k)
38             return arr[pi];
39         else if(pi>k)
40             return quickSelect(arr,low,pi-1,k);
41         else
42             return quickSelect(arr,pi+1,high,k);
43     }
44     return -1;
45 }
```

```

46
47     int main()
48     {
49         printf("CH.SC.U4CSE24221\n");
50
51         int n;
52         scanf("%d",&n);
53
54         int arr[n];
55
56         for(int i=0;i<n;i++)
57             scanf("%d",&arr[i]);
58
59         int k;
60         scanf("%d",&k);
61
62         int result=quickSelect(arr,0,n-1,k-1);
63
64         printf("%d",result);
65
66         return 0;
67     }

```

OUTPUT

```

kantatejesh@KANTAs-MacBook-Air TASK7 % gcc quickselect.c -o quickselect
kantatejesh@KANTAs-MacBook-Air TASK7 % ./quickselect
CH.SC.U4CSE24221
6
7 10 4 5 20 15
5
15
kantatejesh@KANTAs-MacBook-Air TASK7 % 

```



